

Business Analytics – SAS Base

Lecture 5: Character Functions

Aim of the lecture

- Demonstrate how to use character functions in SAS to clean character variables efficiently
- Show you how to manipulate character variables using character functions in SAS

There are a LOT of character functions in SAS

Area	Functions
Remove space	COMPBL, STRIP, COMPRESS, LEFT, TRIM
Concatenate	CAT , II , CATS, CATX
Extract part of the text	SCAN, SUBSTR , LENGTH
Change the appearance	LOWCASE, UPCASE, PROPCASE
Find and replace	INDEX, FIND , TRANWRD, TRANSLATE

This is a list to help you memorize, details will follow

Agenda

- **Character functions – Data cleaning**
- Character functions – Data manipulation
- Recap

You can remove any kind of spaces

Function	Functions
COMPBL	Combines multiple blanks to a single blank
STRIP	Removes leading and trailing spaces
COMPRESS	Removes all spaces
LEFT	Removes leading spaces
TRIM	Removes trailing spaces

Agenda

- Character functions – Data cleaning
- **Character functions – Data manipulation**
- Recap

Concatenation is useful when you need to combine different character variables

Function	Functions
CAT,	Concatenates character strings
CATS	Strips both leading and trailing blanks
CATX	CATX strips both leading and trailing blanks, and insert separators (the first argument to CATX)

SUBSTR and SCAN can be used to extract part of the text

SUBSTR

SUBSTR(text, start_position, number of characters)

Extract strings based on starting position and number of characters

SCAN

SCAN(text, nth word, delimiters)

Extract strings based on words and delimiters

INDEX and FIND can find the location of characters in the text

INDEX

INDEX(text, characters to find)

Returns the location of the characters to find in the text

FIND

FIND(text, characters to find)

Returns the location of the characters to find in the text

From my experience, TRANWRD is more useful

TRANWRD

TRANWRD(text, characters to find, replace with)

Replaces all occurrences of a word in a character string

TRANSLATE

TRANSLATE(text, replace with, characters to find)

Replaces specific characters in a character expression (not by words)

Recap: Now you know how to deal with character strings in SAS

- You can remove any kind of spaces with various functions
- The functions that can manipulate text are more straightforward but need some practices

*Thank
you*

